

Exercise Sheet 2

Discrete Mathematics by Hongfei Fu, 2023.09.14

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1. Show that each of these conditional statements is a tautology.

- a) $[(\neg p) \wedge (p \vee q)] \rightarrow q$
- b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
- c) $[p \wedge (p \rightarrow q)] \rightarrow q$
- d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

Answer Area:

	p	q	$[(\neg p) \wedge (p \vee q)] \rightarrow q$
	T	T	T
a)	F	T	T
	T	F	T
	F	F	T

	p	q	r	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
	T	T	T	T
	T	T	F	T
	T	F	T	T
b)	T	F	F	T
	F	T	T	T
	F	T	F	T
	F	F	T	T
	F	F	F	T

	p	q	$[p \wedge (p \rightarrow q)] \rightarrow q$
	T	T	T
c)	T	T	F
	T	F	T
	T	F	F

	p	q	r	$[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$
	T	T	T	T
	T	T	F	T
	T	F	T	T
d)	T	F	F	T
	F	T	T	T
	F	T	F	T
	F	F	T	T
	F	F	F	T

2. Show that $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent.

Answer Area:

p	q	$\neg(p \leftrightarrow q)$	$p \leftrightarrow \neg q$
T	T	F	F
F	T	T	T
T	F	T	T
F	F	F	F

3. Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent.

Answer Area:

$$\begin{aligned}
 (p \rightarrow q) \vee (p \rightarrow r) &\equiv (\neg p \vee q) \vee (\neg p \vee r) \\
 &\equiv (\neg p \vee \neg p) \vee (q \vee r) \\
 &\equiv \neg p \vee (q \vee r) \\
 &\equiv p \rightarrow (q \vee r)
 \end{aligned}$$

4. Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

Answer Area:

p	q	r	$(p \wedge q) \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	F	F	T	F

In the case above, the two compound propositions don't produce the same truth values, so they are not logically equivalent.

5. Show that the negation $\neg\phi$ of an unsatisfiable compound proposition ϕ is a tautology and the negation $\neg\psi$ of a compound proposition ψ that is a tautology is unsatisfiable.

Answer Area:

- 1) If ϕ is an unsatisfiable compound proposition,
then it is false under any truth assignment for its propositional variables,
then the negation $\neg\phi$ is true under any truth assignment for its propositional variables,
then the negation $\neg\phi$ is a tautology,
- 2) If ϕ is a tautology,
then it is true under any truth assignment for its propositional variables,
then $\neg\phi$ is false under any truth assignment for its propositional variables,
then the negation $\neg\phi$ is unsatisfiable,